

GARNER FIELD, TX, USA

WMO: 722202

Lat: 29.211N

Lon: 99.744W

Elev: 287

StdP: 97.92

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12985

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.4	(c) 0.5	(d) -12.7	(e) 1.3	(f) 7.7	(g) -10.5	(h) 1.6	(i) 8.0	(j) 10.8	(k) 17.0	(l) 9.0	(m) 16.1	(n) 1.6	(o) 50	(p) 0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.9	(c) 38.0	(d) 22.4	(e) 37.4	(f) 22.4	(g) 36.5	(h) 22.5	(i) 25.8	(j) 31.6	(k) 25.3	(l) 31.3	(m) 24.9	(n) 30.8	(o) 4.3	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.9	27.5	23.9	19.4	27.1	23.4	18.8	26.7	81.1	31.7	78.9	31.1	77.1	30.8	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.5	(b) 7.6	(c) 6.9	(e) -5.4	(f) 40.3	(g) 2.1	(h) 1.8	(i) -6.9	(j) 41.6	(k) -8.1	(l) 42.6	(m) -9.3	(n) 43.6	(o) -10.8	(p) 44.9		
(4) 8.5	(5) 7.6	(6) 6.9	(7) -5.4	(8) 40.3	(9) 2.1	(10) 1.8	(11) -6.9	(12) 41.6	(13) -8.1	(14) 42.6	(15) -9.3	(16) 43.6	(17) -10.8	(18) 44.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	11.4	14.1	17.7	21.6	25.1	28.3	29.1	29.5	26.8	22.0	16.2	11.9
	DBStd	7.48	4.61	5.17	4.63	4.04	3.35	2.16	1.69	1.78	2.38	4.19	4.77	4.49
	HDD10.0	102	37	20	4	1	0	0	0	0	0	0	9	31
	HDD18.3	752	218	133	69	15	2	0	0	0	0	18	93	204
	CDD10.0	4180	81	134	243	347	468	549	591	604	503	373	195	91
	CDD18.3	1789	4	14	49	112	212	299	333	346	253	133	29	5
	CDH23.3	19973	46	141	389	1058	2103	3557	4138	4544	2604	1150	200	42
	CDH26.7	10284	6	33	111	429	997	1943	2349	2662	1297	418	37	3
Wind	WSAvg	3.0	2.5	2.9	3.4	3.6	3.7	3.6	3.2	3.1	2.7	2.8	2.5	2.4
Precipitation	PrecAvg	624	27	32	27	58	80	69	47	77	70	72	33	29
	PrecMax	1170	93	98	110	201	200	378	257	402	233	340	79	99
	PrecMin	345	0	0	0	2	3	0	0	0	5	1	0	0
	PrecStd	188	21	28	29	51	51	79	68	88	59	74	26	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	30.8	32.3	36.4	37.5	39.1	39.2	39.0	37.4	34.0	30.9	27.0
		MCWB	15.6	16.8	17.3	19.3	22.0	22.4	22.3	22.5	20.9	22.7	19.7	16.1
	2%	DB	24.3	27.2	30.3	33.7	36.0	37.6	37.8	37.9	36.0	32.3	27.4	24.1
		MCWB	15.4	15.6	18.6	19.9	22.5	22.4	22.5	22.6	22.1	21.7	18.5	16.2
	5%	DB	22.4	24.9	27.4	31.2	34.0	36.5	37.0	37.3	34.5	31.1	25.7	22.5
		MCWB	14.4	15.8	17.5	20.2	22.2	22.5	22.6	22.6	22.4	21.4	18.0	15.7
	10%	DB	20.1	22.6	25.4	29.1	32.1	35.1	35.8	36.4	32.9	29.5	23.8	20.2
		MCWB	14.1	15.4	17.0	19.4	22.3	22.8	22.9	22.8	22.0	21.1	17.5	15.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	20.4	21.8	23.9	25.7	26.6	26.3	25.9	25.8	25.8	22.7	20.8
		MCDB	22.3	24.0	26.1	30.1	31.8	32.0	32.1	32.1	31.5	29.7	25.6	22.7
	2%	WB	18.9	19.5	20.8	23.0	25.0	25.7	25.5	25.3	25.1	24.4	21.6	19.7
		MCDB	21.9	22.5	24.7	28.6	30.8	31.6	31.6	31.7	30.6	28.2	24.3	22.0
	5%	WB	17.0	18.7	20.2	22.3	24.3	25.1	25.0	24.9	24.6	23.5	20.5	18.6
		MCDB	20.0	21.5	23.9	27.2	29.9	31.2	31.1	31.3	29.6	27.3	23.3	20.5
	10%	WB	14.7	17.4	19.5	21.6	23.7	24.6	24.5	24.5	24.1	22.8	19.5	16.5
		MCDB	18.0	20.8	23.0	26.0	29.0	30.5	30.6	30.9	28.8	26.6	22.5	19.1
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.0	13.0	13.4	12.1	12.2	12.5	12.9	11.4	12.6	12.4	12.7
		MCDBR	16.8	16.2	14.7	15.7	14.9	14.4	14.6	14.2	13.6	13.3	13.6	16.1
		MCWBR	9.4	8.6	6.6	6.7	5.1	3.7	3.7	3.8	4.1	5.3	7.3	9.3
	5% WB	MCDBR	11.9	11.4	11.3	11.7	11.4	11.5	12.3	12.7	10.5	10.3	10.6	10.7
		MCWBR	8.4	7.8	6.3	5.5	4.5	3.9	3.6	3.7	3.6	4.9	6.8	8.3
Clear-Sky Solar Irradiance	taub	0.324	0.330	0.351	0.383	0.411	0.410	0.412	0.404	0.394	0.365	0.343	0.321	
	taud	2.497	2.478	2.440	2.373	2.345	2.402	2.422	2.437	2.408	2.481	2.476	2.511	
	Ebn at Noon	916	939	937	911	881	877	874	881	881	890	886	902	
	Edh at Noon	91	101	111	123	127	119	117	114	114	99	92	86	
All-Sky Solar Radiation	RadAvg	3.14	3.78	4.70	5.54	5.82	6.72	6.72	6.45	5.25	4.32	3.32	2.82	
	RadStd	0.37	0.61	0.49	0.56	0.65	0.50	0.57	0.35	0.49	0.60	0.42	0.39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134	+104

Nomenclature: See separate page